

CONGLOMERATE INC'S NEW PDA CASE

Back to 1995

The cellular phone division of Conglomerate Inc. has teamed up with a PC manufacturer to develop, produce, and market a novel hybrid combination of a personal digital assistant (PDA) and a “smart” cellular phone. They have tentatively named it ConneCtor. It transmits and receives both data and voice (unlike competing PDAs which focus on data).

ConneCtor : Another Conglomerate Success Story ?

ConneCtor is lightweight and is shaped like a portable phone with a small backlit LCD touch screen along the handset. Its (open) operating system performs standard cellphone functions and such personal information management (PIM) functions as a calendar, calculator, and address book. It can send and receive faxes, voice messages, and email. Users can input data in four ways :

- By typing on the screen keyboard
- By using a numerical keyboard
- By writing on the screen in “digital ink”
- By speaking into the phone (it includes voice recognition software)

The voice recognition feature is based on a neural network that is trained to recognize a particular user's voice patterns. An additional feature unique to ConneCtor is linkage via wireless local area networks to other PDAs. In summary, the features of Conglomerate's handheld device are :

- Instant communication from PDA to PDA
- Cellular phone and pager, fax and e-mail
- Calendar, scheduler, calculator, and address book
- An open system for customized applications
- A paperless note pad
- Voice recognition

Conglomerate is now trying to identify segments within the market for PDAs, target appropriate segment(s) for ConneCtor, and position ConneCtor in the chosen segments.

Background on the PDA market

In August 1993, Apple introduced its Newton PDA. The broad acceptance that Apple had anticipated did not materialize, and Apple sold only 80,000 Newtons that year. In 1995, the PDA appeared to be on the verge of greater growth and development. The PDA market had grown in four areas : specialized vertical applications (e.g., physician scheduling), PIM (Personal Information Management), mobile communications, and as a supplementary gateway into the Internet. Even though it has four main applications, the PDA is primarily targeted at “road warriors” or “mobile professionals”. This group consists of approximately 25 million people in the United States, of whom about 5 million travel with a computer notebook. Many of these

individuals already have cellular phones and must send and receive a large number of messages and data. The standard PDA cannot handle their needs.



FIGURE 1 – Illustration of a PDA device.

The survey

Conglomerate, Inc. hired a market research firm to survey the market across a broad range of occupation types. The survey includes a screening item asking respondents if they had or would consider a PDA. Only those respondents who answered affirmatively to that question were retained for further analysis.

The questionnaire

The questionnaire asked the respondents to provide data on two kinds of variables : segmentation basis or needs variables and variables that could be used in describing or targeting the clusters.

Questions for determining segmentation

- X1 Whenever new technologies emerge in my field, I am among the first to adopt them.
(1 = Strongly disagree, 7 = Strongly agree)
- How often do you use the following : (1 = Never, 7 = Always)
- X2 a. Pager ?
- X3 b. Phone or voice mail ?
- X4 c. Scheduling or contact-management tools, i.e., filofax or similar devices ?
- X5 How often do others send you time-sensitive information (e.g., work orders)? (1 = Never, 7 = Daily)
- X6 How often do you have to send time-sensitive information while away from your office ? (1 = Never, 7 = Daily)
- X7 How much of your time do you spend away from your office location ? (1 = 0%, 7 = 70% or more)
- X8 How important is wireless communication to you ? (1 = Not at all important, 7 = Very important)
- X9 How important is it for you to share information rapidly with colleagues while away from an office location ? (1 = Not at all important, 7 = Very important)

How much would you be willing to pay for a personal digital assistant (PDA) with the following features : instant communication from PDA to PDA, cellular phone and pager, fax and email, calendar, scheduler, calculator, address book, open system for customized applications, paperless note pad, and voice recognition ?

- X10 a. Monthly (for all services that you use) ?
- X11 b. Invoice price for the PDA device with all features ?

Questions for describing or targeting the clusters

- Z1 Age
- Z2 Education (1 = High school, 2 = Some college, 3 = College, 4 = Graduate degree)
- Z3 Income (in thousands of dollars per year)
- Z4 Construction (0 = No, 1 = Yes)
- Z5 Emergency (fire, police, ambulance, etc.) (0/1)
- Z6 Sales (insurance, pharmacy, etc.) (0/1)
- Z7 Maintenance and service (0/1)
- Z8 Professional (e.g., lawyer, consultant, etc.) (0/1)
- Z9 Do you own a PDA ? (0/1)
- Media consumption (Readership of magazines) : (0 = No, 1 = Yes)
- Z10 Businessweek
- Z11 PC Magazine
- Z12 Field & Stream
- Z13 Modern Gourmet

EXERCISES

1. Run only cluster analysis on the data to try to identify the number of distinct segments present in this market. Consider both the distances separating the segments and the characteristics of the resulting segments.
2. Identify and profile (name) the clusters that you select. Given the attributes of ConneCtor, which cluster would you target for your marketing campaign ?
3. Describe the segments using the answers of the second set of questions. How would you go about targeting the segment(s) you picked in question 2 ?
4. How has this analysis helped you to segment the market for ConneCtor ?

HAC Applications – Correction

Case 1 : “DEMO” Data Sheet (Countries)

The DEMO file contains an illustrative data set of $n = 18$ countries, each described by four quantitative variables (for example : growth rate of population, GDP per capita, life expectancy, fertility index).

1.1 Impact of a Germany–Bulgaria Merge on the Classification Quality

We consider k groups $G_1, \dots, G_k$ and the merge of two groups G_i and G_j into a single group $G_i \cup G_j$. Under Ward’s criterion, the loss of inter-class inertia (that is, the decrease in the classification quality indicator) is given by

$$\Delta I_{\text{inter}} = I(G_1, \dots, G_i, \dots, G_j, \dots, G_k) - I(G_1, \dots, G_i \cup G_j, \dots, G_k) = \frac{n_i n_j}{n(n_i + n_j)} d^2(g_i, g_j), \quad (1)$$

where

- n is the total number of individuals,
- n_i and n_j are the sizes of G_i and G_j ,
- g_i and g_j are the centroids (barycentres) of G_i and G_j ,
- $d^2(g_i, g_j)$ is the squared Euclidean distance between g_i and g_j .

In the DEMO file, each country is a single individual. If we consider Germany (DE) and Bulgaria (BG) as two one-element groups, we have

$$n_{\text{DE}} = n_{\text{BG}} = 1, \quad n = 18.$$

By plugging $n_i = n_j = 1$ into the formula, the impact on the classification quality indicator when merging Germany and Bulgaria is

$$\Delta I_{\text{inter}}(\text{DE}, \text{BG}) = \frac{1 \cdot 1}{18(1 + 1)} d^2(g_{\text{DE}}, g_{\text{BG}}) = \frac{1}{36} d^2(g_{\text{DE}}, g_{\text{BG}}). \quad (2)$$

The squared distance $d^2(g_{\text{DE}}, g_{\text{BG}})$ is computed using the four original variables (for example : growth rate, GDP per capita, life expectancy, fertility index).

The larger $\Delta I_{\text{inter}}(\text{DE}, \text{BG})$ is, the larger the deterioration of the classification quality if we force Germany and Bulgaria to be in the same cluster.

1.2 Variable Importance and Need for Normalization

The DEMO variables are not on the same scale. In particular, GDP per capita is measured in thousands of dollars and has much larger numerical values than the other variables. In the Euclidean distance

$$d^2(g_{\text{DE}}, g_{\text{BG}}) = \sum_{\ell} (x_{\text{DE}, \ell} - x_{\text{BG}, \ell})^2,$$

the term corresponding to GDP per capita dominates the sum. As a consequence, the Ward criterion and the resulting hierarchy will essentially reflect differences in GDP per capita, while the other dimensions (demography, health, fertility) will be under-weighted.

Proposed solution. To give each variable a comparable influence, we should normalize the variables before running the HAC :

- center and reduce each variable (z-scores)

$$z_\ell = \frac{x_\ell - \bar{x}_\ell}{s_\ell},$$

where $\bar{x}_\ell$ is the sample mean and s_ℓ the sample standard deviation ;

- perform HAC (Ward) on the standardized data.

After this normalization, each variable contributes on a similar scale to the distance and to the Ward criterion.

1.3 HAC With and Without Normalization

HAC on raw data. On the raw data, Ward’s method tends to group countries mainly according to their level of GDP per capita : the richer countries form one or several clusters, poorer countries form others. Some merges, such as a Germany–Bulgaria merge, are driven almost entirely by the economic variable.

HAC on normalized data. On the standardized data, the first merges reflect a more balanced compromise between the different dimensions :

- economic level (GDP per capita),
- demographic dynamics (growth rate, fertility),
- health (life expectancy).

The resulting clusters are therefore easier to interpret as “profiles” of countries combining several aspects, rather than pure “rich/poor” groups.

1.4 Comparison with k -means

If we choose a number of clusters k and run k -means on the same standardized DEMO data, we obtain a partition that is generally very similar to the one obtained by cutting the Ward dendrogram at level k :

- both methods optimize a within-cluster inertia criterion ;
- HAC (Ward) does so in a hierarchical way, without being able to undo previous merges ;
- k -means directly partitions the data into k groups, with results that may depend on initialization.

On a small data set such as DEMO, the two approaches typically provide consistent country groupings once the variables are standardized.

Case 2 : Conglomerate Inc.’s New PDA (ConneCtor)

We now analyze the PDA survey. Each individual is described by :

- needs variables X_1 to X_{11} (Innovator, Use_Pager, Use_Phone, Use_Schdl, Inf_Passive, Inf_Active, Remote_Loc, Wireless, Share_Inf, Monthly, Invoice),
- descriptive/targeting variables Z_1 to Z_{13} (age, education, income, type of occupation, PDA ownership, magazine readership).

2.1 Number of Segments

We perform a hierarchical agglomerative clustering with Ward’s method on the standardized needs variables (X_1 to X_{11}). Inspection of the dendrogram and of the successive jumps in the

Ward criterion suggests that a cut at

$$k = 3$$

segments provides a good compromise :

- the partition into 3 clusters avoids merging clearly distinct groups ;
- going beyond 3 clusters produces very small groups that are less stable and more difficult to interpret.

We therefore retain *three* market segments.

2.2 Cluster Profiling (Needs Variables) and Target Choice

Using the R output (see Appendix), we obtain three clusters with distinct profiles on the X variables.

Segment A : “Advanced tech professionals” This segment corresponds to the cluster with :

- the highest average score on *Innovator* (around 5.9 on a 1–7 scale) ;
- relatively frequent use of phone/voice mail and scheduling tools ;
- strong use of time-sensitive information (sending/receiving) ;
- a relatively high invoice price they are ready to pay for the device ;
- standardized usage levels that indicate a global affinity with new technologies.

These individuals are technologically inclined and value powerful tools that help them manage information and communication.

Segment B : “Mobile field and service workers” This segment has :

- moderate innovativeness scores (neither very low nor very high) ;
- frequent use of pager and time-sensitive communications ;
- a high proportion of time spent away from the office (Remote_Loc) ;
- strong importance attached to wireless communication and rapid information sharing (Wireless and Share_Inf are high) ;
- a relatively high monthly amount they are willing to pay for the service bundle.

These individuals are often “on the road”, need to stay in touch with their colleagues and headquarters, and clearly perceive the value of wireless connectivity.

Segment C : “Light or conservative users” This segment is characterized by :

- low scores on *Innovator* ;
- low to moderate usage of advanced tools (pager, scheduling) ;
- some use of phone/voice mail but less interest in wireless and information sharing ;
- low willingness to pay, both monthly and for the device.

They do not have strong needs for an advanced PDA/phone hybrid and are unlikely to adopt ConneCtor in the short term.

Target choice. ConneCtor offers :

- advanced wireless communication (PDA to PDA, LAN),
- full integration of phone, pager, fax, email,
- scheduling and information management,
- voice recognition and an open operating system.

These features are particularly valuable for users who :

- often work away from the office,
- need to send and receive time-sensitive information,
- are ready to invest in a high-end, multi-function device.

Segments A and B are therefore the most relevant :

- Segment A brings high income, high education, and strong familiarity with technology ;
- Segment B expresses the strongest need for wireless and mobile communication.

In practice, the **primary target** can be defined as the intersection of these two logics : mobile professionals with high technological affinity and a strong need for wireless communication. From a clustering point of view, this corresponds mainly to the *advanced tech professionals* (Segment A), supplemented by part of the *mobile field and service workers* (Segment B).

2.3 Using Descriptive Variables to Describe and Reach the Segments

The descriptive variables Z_1 to Z_{13} show that :

- Segment A tends to be :
 - slightly younger to middle-aged adults (mid-30s),
 - with higher education (college or graduate degree),
 - with higher income,
 - over-represented in professional occupations (consultants, lawyers, engineers, etc.),
 - frequent PDA owners,
 - frequent readers of business and technology magazines (BusinessWeek, PC Magazine).
- Segment B includes more people in construction, emergency, maintenance, and service jobs, who :
 - work a large share of their time outside the office,
 - have strong operational communication needs,
 - are particularly sensitive to wireless and PDA-to-PDA communication.
- Segment C is more heterogeneous but includes many sales-oriented or traditional users with lower income and lower technology appetite.

Targeting the chosen segment(s). To reach the selected segments (primarily A, secondarily the most mobile part of B), Conglomerate Inc. can :

- advertise in business and technology media (BusinessWeek, PC Magazine, specialized IT and telecommunications press) ;
- use direct marketing via companies employing mobile professionals (consultancies, law firms, engineering companies, emergency and service organizations) ;
- establish partnerships with cellular operators and notebook manufacturers to sell ConneCtor as a premium option for “road warriors” ;
- participate in professional trade fairs where advanced mobile communication tools are presented.

2.4 Contribution of the Analysis to ConneCtor’s Market Segmentation

The HAC on the PDA data contributes to Conglomerate Inc.’s marketing decisions in several ways :

- It shows that the PDA market is not homogeneous : some users have strong, complex needs and a high willingness to pay, while others are light users with limited interest in advanced features.
- It provides quantitative profiles of the segments on the needs variables (X), which clarifies which features of ConneCtor (wireless communication, PDA-to-PDA link, voice recognition) are most valued and by whom.
- It links these segments to observable characteristics (Z) : age, income, occupation, media consumption, current PDA ownership. This makes the segments operational for targeting.

- It offers a clear rationale for positioning : ConneCtor should be positioned as a high-end, professional tool for mobile, tech-oriented users, rather than as a mass-market gadget for all potential PDA owners.

Overall, the analysis helps Conglomerate Inc. to focus its marketing resources on the segments for which ConneCtor creates the most value, and for which the probability of adoption is highest.

Appendix : R code for HAC applications

Case 1 : DEMO data (countries)

```
# Case 1: DEMO data (countries) - HAC and impact of DE-BG

# Clear workspace
rm(list = ls())

# Load DEMO data (one row per country, variables such as GDP, life expectancy...)
# Adjust the file name / path if needed
demo <- read.csv("demo.csv", header = TRUE, stringsAsFactors = FALSE)

# Convert numeric columns with commas to proper numeric values
demo$T.Ch <- as.numeric(gsub(",", ".", demo$T.Ch))
demo$PNB.H <- as.numeric(gsub(",", ".", demo$PNB.H))
demo$EspV <- as.numeric(gsub(",", ".", demo$EspV))
demo$IndFec <- as.numeric(gsub(",", ".", demo$IndFec))

row.names(demo) <- demo$Pays

# Number of countries
n <- nrow(demo)

# 1.1 Impact of merging Germany (DE) and Bulgaria (BG)

de <- demo["Allemagne", c("T.Ch", "PNB.H", "EspV", "IndFec")]
bg <- demo["Bulgarie", c("T.Ch", "PNB.H", "EspV", "IndFec")]

# Squared Euclidean distance between DE and BG
de_bg_dist2 <- sum((de - bg)^2)

# Ward impact on inter-class inertia:
# Delta_I_inter(DE,BG) = (n_i * n_j) / (n * (n_i + n_j)) * d^2(g_i, g_j)
# Here n_DE = n_BG = 1
delta_I_inter_DE_BG <- (1 * 1) / (n * (1 + 1)) * de_bg_dist2
print(delta_I_inter_DE_BG)

# 1.2 HAC on raw data (no normalization)

dist_demo_raw <- dist(demo[, c("T.Ch", "PNB.H", "EspV", "IndFec")])
hac_demo_raw <- hclust(dist_demo_raw, method = "ward.D2")
plot(hac_demo_raw, main = "HAC on DEMO (raw data)")
```

```

# 1.3 HAC on standardized data (normalization)

demo_std      <- scale(demo[, c("T.Ch", "PNB.H", "EspV", "IndFec")])
dist_demo_std <- dist(demo_std)
hac_demo_std  <- hclust(dist_demo_std, method = "ward.D2")
plot(hac_demo_std, main = "HAC on DEMO (standardized data)")

# 1.4 Comparison with k-means (on standardized data)

set.seed(123)
k <- 3

km_demo <- kmeans(demo_std, centers = k, nstart = 50)

# Cluster sizes
km_demo$size

# Cluster means (on standardized variables)
km_demo$centers

# Compare with a cut of the HAC tree at k clusters
cut_hac_demo <- cutree(hac_demo_std, k = k)
table(cut_hac_demo)

```

Case 2 : Conglomerate Inc.'s new PDA (ConneCtor)

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# Case 2: PDA data - HAC and segment profiling (3 clusters)

# Clear workspace
rm(list = ls())

# Load PDA data (Id, X1-X11 = needs, Z1-Z13 = descriptors)
pda <- read.csv("pda.csv", header = TRUE, stringsAsFactors = FALSE)

# Clean column names (remove leading/trailing spaces)
names(pda) <- trimws(names(pda))

# Extract needs variables X1-X11
X <- pda[, c("Innovator", "Use_Pager", "Use_Phone", "Use_Schdl",
"Inf_Passive", "Inf_Active", "Remote_Loc", "Wireless",
"Share_Inf", "Monthly", "Invoice")]

# Convert Monthly to numeric and impute missing values by the mean
X$Monthly <- as.numeric(trimws(X$Monthly))
X$Monthly[is.na(X$Monthly)] <- mean(X$Monthly, na.rm = TRUE)

# Standardize needs variables

```

```

X_std <- scale(X)

# HAC (Ward) on standardized needs variables
dist_pda_std <- dist(X_std)
hac_pda_std <- hclust(dist_pda_std, method = "ward.D2")

plot(hac_pda_std, main = "HAC on PDA (standardized X1-X11)")

# Cut the tree into 3 segments
k <- 3
seg <- cutree(hac_pda_std, k = k)

# Segment sizes
table(seg)

# Attach segment membership to original data
pda$seg <- seg

# 2.3 Cluster profiling: means of needs variables (X1-X11)
X_means_by_seg <- aggregate(X, by = list(segment = seg), FUN = mean)
print(X_means_by_seg)

# 2.4 Cluster profiling: means of descriptive variables (Z1-Z13)
Z <- pda[, c("Age", "Education", "Income", "Construct", "Emergency",
"Sales", "Service", "Professnl", "PDA", "Bus_Week",
"PC_Mag", "Field&Stream", "M_Gourmet")]

Z_means_by_seg <- aggregate(Z, by = list(segment = seg), FUN = mean)
print(Z_means_by_seg)

# Optionally save results
write.csv(X_means_by_seg, file = "pda_X_means_by_segment.csv",
row.names = FALSE)
write.csv(Z_means_by_seg, file = "pda_Z_means_by_segment.csv",
row.names = FALSE)

write.csv(pda, file = "pda_with_segments.csv", row.names = FALSE)

```